

Comparative Analysis of Automated US Housing Valuation Models

1. Motivation and Research Question

Real estate constitutes a key component in collateral within the financial system. Accurate property valuations are essential to maintain its efficiency and stability. However, house price shows heterogeneity across regions, and the impacts of property characteristics on price demonstrate nonlinearity and cross interactions, which imposes challenges on traditional approaches. The core question remains how accurately we can predict house prices by only utilizing observable structural characteristics and coarse-grained location data. Besides, it is also worth exploring that whether machine learning methods significantly improve the prediction accuracy compared to hedonic linear model.

2. Literature Review

The traditional theoretical foundation of housing valuation is based on the hedonic pricing theory, within which property is viewed as a combination of attributes and the price reflects the implied value of those characteristics (Rosen, 1974). However, even though traditional hedonic models have strong interpretability, they typically rely on strong assumptions and may face severe limitations when encountering nonlinearity and complexity. Recent research on automated valuation model increasingly employs machine learning approaches to enhance the accuracy of valuation. Among these methods, tree-based models, especially gradient boosting tree, have shown strong capability in prediction by flexibly capturing nonlinearity correlations and high dimensional interactions (Hjort et al., 2022). Research by Sharma et al. (2024) also supports this idea. Built on these literatures, the project compares OLS with fixed effects, LASSO, and XGBoost to investigate the performance in US real estate housing valuation.

3. Data

This project uses the USA Real Estate Dataset with around 1.5 million observations obtained from Kaggle. Even though richer information like community quality or school districts are not included, the dataset contains listing price with property structural and location characteristics essential for valuation. With target variable “price”, there is no missing value and data type error. Variables “brokered by” and “street” are not included since they contain mostly unique value for identification, and observations with price under 10,000 are dropped since purely unrealistic. Continuous variables are winsorized at 1st and 99th percentile to mitigate the influence of outliers, followed by a log transformation to address the right skewness. Besides, discrete numeric variables are also capped by the 99.9th percentile. For “state” and “state”, those ranked 200th and 52nd or below has been grouped under “other”, respectively, while “area type” remains unchanged. The data is then saved into a parquet file for model training after cleaning.

4. Methodology

To compare the performance of various models in housing valuation, the project adopted OLS with fixed effects, LASSO, and XGBoost, which represents traditional linear prediction, regularized linear machine learning method, and flexible nonlinear tree model, respectively. OLS with fixed effects provides traditional baseline predictions, it utilized property characteristics as core explanatory variables and controlled for location fixed effects like state, city, and area type. It is most interpretable while it may face challenges in dealing with nonlinearity in realistic scenarios. Additionally, LASSO is employed to examine whether regularization may improve prediction performance. It imposes a L1 penalty term to high dimensional encoded categorical variables to achieve variables selection and parameters shrinkage, maintaining the model stability. However, its improvement does not reflect in functional flexibility. Most importantly, XGBoost is adopted as the main model. It bases on the boosting framework to automatically capture the nonlinearity and cross interactions among variables by a combination of multiple trees. Theoretically, XGBoost is suitable for housing valuation as there may be significant heterogeneity among property characteristics.

5. Implementation Details

In model realization, data is split into “train set”, “validation set” and “test set”, with a proportion of 72%, 8% and 20%. The train set is the main set for data training and the test set is out of sample for model evaluation, while the validation set is mainly used for early stopping. In variables handling, columns “price per acre” and “price per square feet” are excluded from any pipelines as they may contain strong information of the target variable. OLS incorporates the variables into formular setting for regressions, while LASSO and XGBoost will one-hot encode categorical variables before the pipeline. Numeric variables remain unchanged. LASSO with cross validation chooses regularized parameter α , and a large α shrinks many coefficients to 0 while LASSO with small α is like OLS. The training set for both LASSO and OLS is the union of train set and validation set as there is no need for early stopping.

In terms of XGBoost, it aims to tune and get the best hyperparameters, including learning rate, max depth, and minimum child weight, in a smaller random selected set for computational efficiency. Cross validation and 1500 estimators are included in the tuning process. After getting the best hyperparameters, they are applied to the whole train set for training, which uses much more estimators with a validation set to achieve early stopping.

RMSE, MAE and R^2 are employed to evaluate model performance. The sensitivity of RMSE to large prediction errors makes it suitable to evaluate severe misestimation, while MAE directly reflects the average deviation from the sample. R^2 simply measures the fit of models.

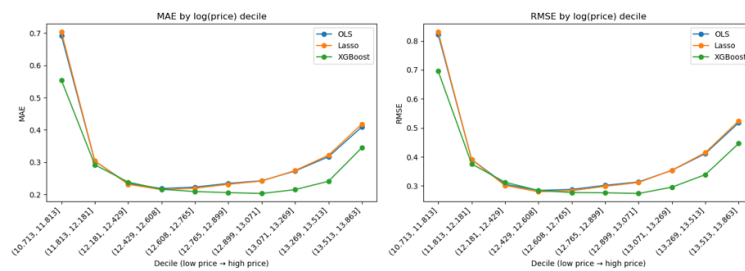
6. Results

Three models demonstrate great disparity in performance. In general, OLS slightly outperforms LASSO while XGBoost is much better than the other two by all metrics. It indicates the

limitation of linear models lie in the functional forms rather than overfitting from high dimensional variables. Since location variables contain valuable information in house valuation, shrinkage does not necessarily enhance out of sample performance, but brings additional bias to the model, leading to the worse performance of LASSO. In contrast, the dominant advantages in XGBoost indicates that the correlation between house price and its characteristics is not simply linear, and these nonlinearity and cross interactions are captured by tree-based model effectively, leading to the best out of sample performance among the three models.

Lasso - Model Evaluation	OLS - Model Evaluation	XGBoost - Model Evaluation
RMSE: 0.4308, MAE: 0.3165, R2: 0.5710	RMSE: 0.4293, MAE: 0.3151, R2: 0.5740	RMSE: 0.3791, MAE: 0.2724, R2: 0.6678

The plot compares the MAE and RMSE of three models across segmented price. Within the models, the U-shape curves indicate that estimation errors in tails are much higher than those in the middle, implying that properties in tails show strong heterogeneity and are difficult to value accurately. Among the models, XGBoost effectively decrease the estimation errors in tails and demonstrates a dominant capability in most deciles.



As for variables, OLS and LASSO regressions (Appendixes 1, 2) show that house size, bath, and acre lot are positive correlated with house price, while bed is negative correlated, possibly indicating that more bedrooms may reflect a tighter layout conditional on other variables. Meanwhile, partial dependence plot from XGBoost (Appendix 3) indicates that correlations between variables and price are nonlinear for all numeric variables, which helps explain how capturing nonlinearity may outperform linear models out of sample in housing valuation.

7. Discussion

The project finds out that XGBoost performs the best in predicting US housing valuation while LASSO does not improve the results from OLS, implying that there exists complicated nonlinearity between house price and property characteristics that linear model (with regularization) cannot fully capture. Besides, the prediction errors remain large in tails, indicating that current characteristics are not enough to mark down the pricing mechanisms for low and high price houses, which points out the direction for future improvement by incorporating detailed-oriented variables like community quality, crime rate, and transportation viability, and potentially using price segment specific for modelling.

Appendixes

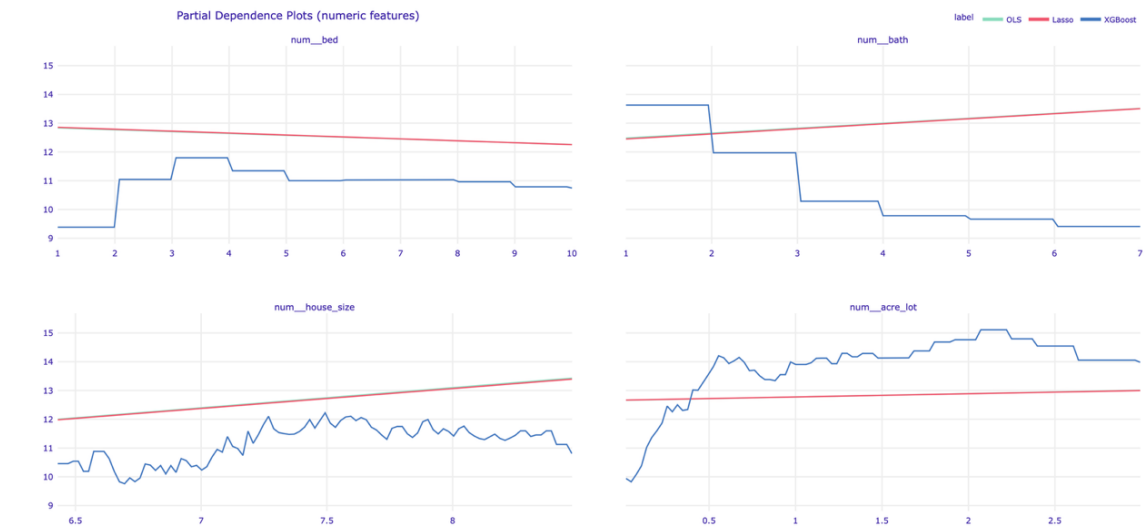
1. OLS Regression Results

OLS Regression Results					
Dep. Variable:	price	R-squared:	0.574		
Model:	OLS	Adj. R-squared:	0.574		
Method:	Least Squares	F-statistic:	6197.		
Date:	Sat, 07 Mar 2026	Prob (F-statistic):	0.00		
Time:	16:48:50	Log-Likelihood:	-6.7397e+05		
No. Observations:	1176348	AIC:	1.348e+06		
Df Residuals:	1176091	BIC:	1.352e+06		
Df Model:	256				
Covariance Type:	nonrobust				
	Coef	Std Err	t	P> t	[0.025 0.975]
Intercept	6.5567	0.0156	420.0319	0.0000	6.5262 6.5873
bed	-0.0642	0.0006	-116.4748	0.0000	-0.0653 -0.0631
bath	0.1720	0.0007	264.4456	0.0000	0.1707 0.1733
house_size	0.6996	0.0017	423.6193	0.0000	0.6964 0.7028
acre_lot	0.1155	0.0016	73.2469	0.0000	0.1124 0.1186
+ 199 city, 51 state, 2 area_type dummies (not shown)					

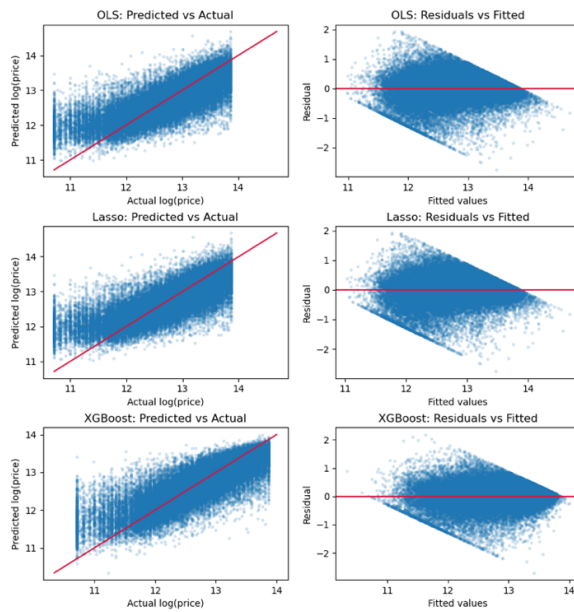
2. LASSO Regression Results

Tuning & training Lasso regression...	
Fitting 3 folds for each of 10 candidates, totalling 30 fits	
Best parameters found: {'alpha': 0.0001}	
Best score found: -0.1855723461019034	
Lasso model trained successfully	
Lasso trained (alpha=0.000100, time=15.3s)	
=====	
Lasso - Model Evaluation	
=====	
RMSE: 0.4308, MAE: 0.3165, R2: 0.5710	
=====	
Lasso Regression Summary	
=====	
Observations: 1,176,348 Features: 259	
Alpha (λ): 0.000100	
Non-zero features: 188 / 259 (72.6%)	
R²: 0.571133 Adj R²: 0.571065	
Residual Std Err: 0.430745	
=====	
Coef	
=====	
const	7.1977
bed	-0.0668
bath	0.1759
house_size	0.6919
acre_lot	0.1110
area_type_rural	-0.0529
area_type_suburban	0.0000 (zeroed)
area_type_urban	0.0047
=====	
City dummies kept: 137 / 200	
State dummies kept: 45 / 52	
Note: Lasso performs variable selection; SEs/p-values not reported.	

3. Partial Dependence Plots (PDPs)



4. Model Comparison in terms of fitness



5. Variables Importance



Reference:

1. Hjort, A., Pensar, J., Scheel, I., & Sommervoll, D. E. (2022). House price prediction with gradient boosted trees under different loss functions. *Journal of Property Research*, 39(4), 338-364.
2. Rosen, S. (1974). Hedonic prices and implicit markets: product differentiation in pure competition. *Journal of political economy*, 82(1), 34-55.
3. Sharma, H., Harsora, H., & Ogunleye, B. (2024). An optimal house price prediction algorithm: XGBoost. *Analytics*, 3(1), 30-45.